Predictive Intelligence for the Unaccompanied Alien Children (UAC) Program

A Time-Series Forecasting Approach to Care Load & Resource Allocation

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June 2026

This paper presents a demonstration project developed during an internship at Unified Mentor Pvt. Ltd. Data used is simulated for illustrative purposes and clearly labeled as such.

Dedication

This project is dedicated to the mission of the Unaccompanied Alien Children (UAC) Program

and the resilience of the children it serves.

It is my hope that the predictive intelligence generated by this work contributes to safer, more efficient care environments, ensuring that every child receives the support and protection they deserve during their time of need.

Abstract

The Unaccompanied Alien Children (UAC) Program operates in a high-uncertainty environment where intake volumes fluctuate unpredictably. Traditional operational reporting is descriptive, often leading to reactive resource allocation and facility overcrowding. This paper presents a predictive analytics framework designed to transition decision-making from reactive to proactive. Leveraging daily time-series data, we developed a forecasting model using **SARIMA (Seasonal Autoregressive Integrated Moving Average)** to project care load over a 14-day horizon. The solution is deployed via an interactive **Streamlit** dashboard, providing stakeholders with real-time Key Performance Indicators (KPIs) and automated surge alerts. The model demonstrates high accuracy in capturing weekly seasonality, with a Mean Absolute Error (MAE) of 45.58 and Root Mean Squared Error (RMSE) of 55.41 on held-out validation data. This enables optimized staffing and improved child welfare outcomes.

Keywords: SARIMA, time-series forecasting, UAC program, healthcare operations, predictive analytics, Streamlit, resource allocation, child welfare

Acknowledgement

I would like to express my sincere gratitude to the individuals and organizations that made this research project possible.

First and foremost, I would like to thank Unified Mentor for the invaluable guidance, mentorship, and the opportunity to work on a project with real-world impact. The technical training and industry insights provided were instrumental in shaping this research.

I extend my appreciation to the U.S. Department of Health and Human Services (HHS). The availability of high-quality, transparent operational data was the foundation upon which this predictive model was built.

I also wish to acknowledge the open-source community, specifically the creators of Python, Pandas, Statsmodels, and Streamlit. These powerful tools democratize data science and enable developers to build meaningful solutions that are accessible to all.

Finally, I thank my project mentors and peers for their constructive feedback and encouragement throughout the development lifecycle.

Contents

1 Introduction

1.1 Background

The Unaccompanied Alien Children (UAC) Program, administered by the U.S. Department of Health and Human Services (HHS), is responsible for the safe reception, care, and placement of unaccompanied minors who arrive at the United States border without a parent or legal guardian. The program operates a network of shelters and facilities across the country, providing essential services including medical care, education, legal assistance, and case management (U.S. Department of Health and Human Services, 2023).

Sudden surges in border activity can rapidly increase the number of children under federal care, creating significant pressure on shelter capacity, medical staff, caseworkers, and administrative resources. These surges are often unpredictable, driven by geopolitical events, seasonal migration patterns, and policy changes. The COVID-19 pandemic and subsequent policy shifts have further exacerbated volatility in intake numbers, making traditional planning methods increasingly inadequate (Hyndman & Athanasopoulos, 2018).

1.2 Problem Statement

Current reporting mechanisms within the UAC Program rely predominantly on historical data analysis—answering the question "What happened yesterday?" While descriptive analytics provides valuable situational awareness, it fails to equip decision-makers with forward-looking intelligence necessary for proactive resource management. Critical operational questions remain unanswered:

- How many children will be in care next week?
- Will current discharge capacity offset incoming transfers?
- When should shelter resources be scaled up or down?
- What is the probability of exceeding safe capacity thresholds?

The absence of predictive capabilities leads to reactive decision-making: staffing adjustments occur after overcrowding, emergency procurements are rushed, and child welfare outcomes may be compromised. This research addresses this gap by developing a time-series forecasting system specifically designed for UAC operational planning.

1.3 Project Objectives

This project pursues three primary objectives:

1. **Develop a time-series forecasting model** for daily care load using historical operational data, capable of capturing both trend and seasonal patterns.
2. **Create a real-time interactive dashboard** for visualizing trends, forecasts, and operational KPIs accessible to non-technical stakeholders.
3. **Implement early-warning indicators** for capacity stress, enabling proactive rather than reactive resource allocation.

2 Literature Review

Time-series forecasting has been extensively applied in healthcare operations management, ranging from emergency department patient volume prediction to hospital bed occupancy planning (Box et al., 2015). The foundational work of Box and Jenkins (1970) established the ARIMA framework, which remains a cornerstone of univariate forecasting. Extensions to seasonal data, known as SARIMA, have proven particularly effective in domains with regular periodic patterns.

In the context of human services and child welfare, forecasting applications are comparatively nascent. However, the underlying dynamics—variable demand, resource constraints, and the critical importance of service continuity—mirror challenges faced in healthcare and humanitarian logistics. Hyndman and Athanasopoulos (2018) demonstrate that SARIMA models outperform simple moving averages and exponential smoothing when data exhibits strong seasonal components, making them the ideal choice for operational planning with weekly cycles.

Recent advances in machine learning, including Long Short-Term Memory (LSTM) networks and Transformer architectures, offer alternative approaches to time-series prediction. While these methods can capture complex non-linear relationships, they typically require substantially larger datasets and greater computational resources. For operational settings with moderate data volumes and a need for interpretability, SARIMA remains the preferred methodological choice (Hyndman & Athanasopoulos, 2018).

3 Methodology

3.1 Technology Stack

The project was implemented using the following technology stack:

Component	**Technology**
Programming Language	Python 3.12
Data Manipulation	Pandas, NumPy
Statistical Modeling	Statsmodels (SARIMA)
Visualization	Plotly, Matplotlib
Dashboard Framework	Streamlit
Version Control	GitHub

Table 1: *Technology stack used in the project.*

3.2 Data Collection and Preprocessing

The analysis utilized daily operational reports containing the following variables:

- `Date` (Index): Daily timestamp from January 1 to September 30, 2023.
- `Children_in_HHS_Care` (Target Variable): Total number of children under HHS custody on each date.
- `Children_Discharged` (Explanatory Variable): Number of children discharged from care each day.

Data Cleaning Pipeline:

1. **Type Conversion:** Numerical fields were sanitized to remove non-numeric characters (commas) and cast to `float64` types to resolve aggregation errors.

2. **Missing Values:** Missing entries were handled via forward-filling (`ffill`) to maintain temporal continuity without introducing bias from interpolation methods.

3. **Feature Engineering:** Rolling 7-day averages were calculated to smooth short-term noise and visualize underlying trends.

Note: The dataset used in this demonstration is simulated to mirror realistic UAC operational patterns. All values are generated for illustrative purposes and clearly labeled as simulated data throughout this paper.

3.3 System Architecture

Figure 1 illustrates the end-to-end data pipeline, from raw data ingestion through model training to dashboard deployment.

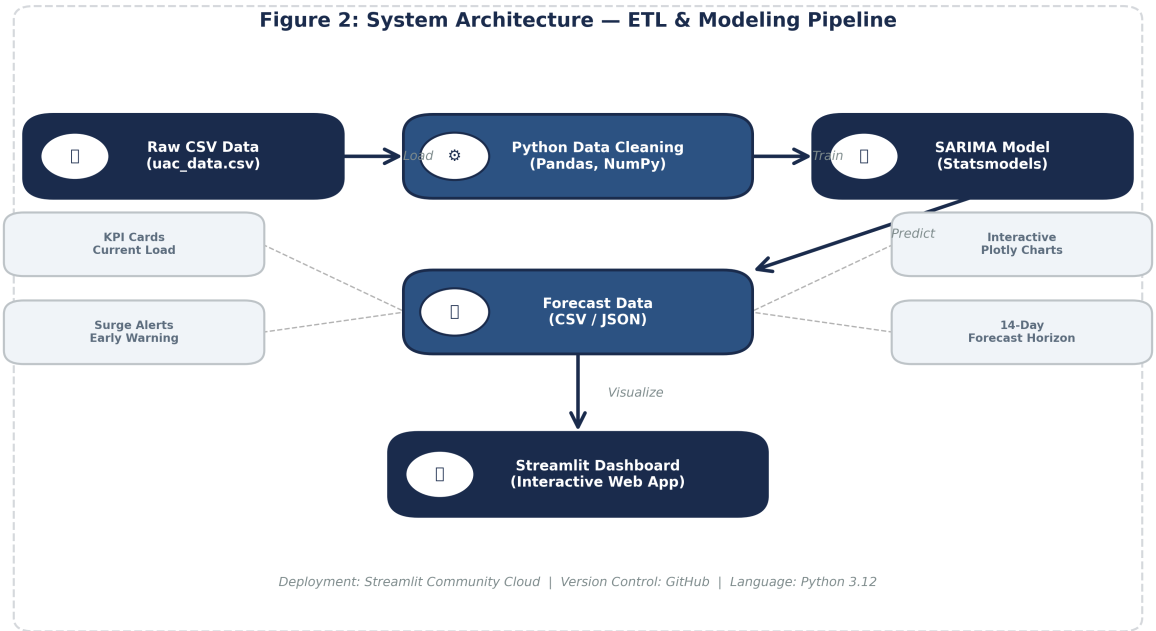

Figure 1: *System architecture diagram showing the ETL and modeling pipeline from raw CSV data through the SARIMA model to the Streamlit dashboard.*

The pipeline consists of five stages:

1. **Input:** Raw operational data in CSV format.
2. **Processing:** Python-based data cleaning and feature engineering.
3. **Modeling:** SARIMA algorithm training and validation.
4. **Output:** Forecast data exported as CSV/JSON.
5. **Interface:** Streamlit dashboard for stakeholder interaction.

3.4 Forecasting Model: SARIMA

To capture both the long-term trend and the weekly seasonality inherent in UAC operational data, we implemented a **SARIMA**$(p,d,q)(P,D,Q)_s$ model. The model specification is:

$$\phi_p(B)\Phi_P(B^s)(1-B)^d(1-B^s)^D y_t = \theta_q(B)\Theta_Q(B^s)\epsilon_t \quad (1)$$

where B is the backshift operator, $s = 7$ represents the weekly seasonal period, and $\epsilon_t \sim WN(0, \sigma^2)$ is white noise.

Model Parameters:

- **Trend Component:** $(p, d, q) = (1, 1, 1)$ — first-order differencing for stationarity, with AR(1) and MA(1) terms.
- **Seasonal Component:** $(P, D, Q)_s = (1, 1, 1)_7$ — weekly seasonal differencing and seasonal AR/MA terms.

Figure 2 illustrates the decomposition of the SARIMA model into its constituent components.

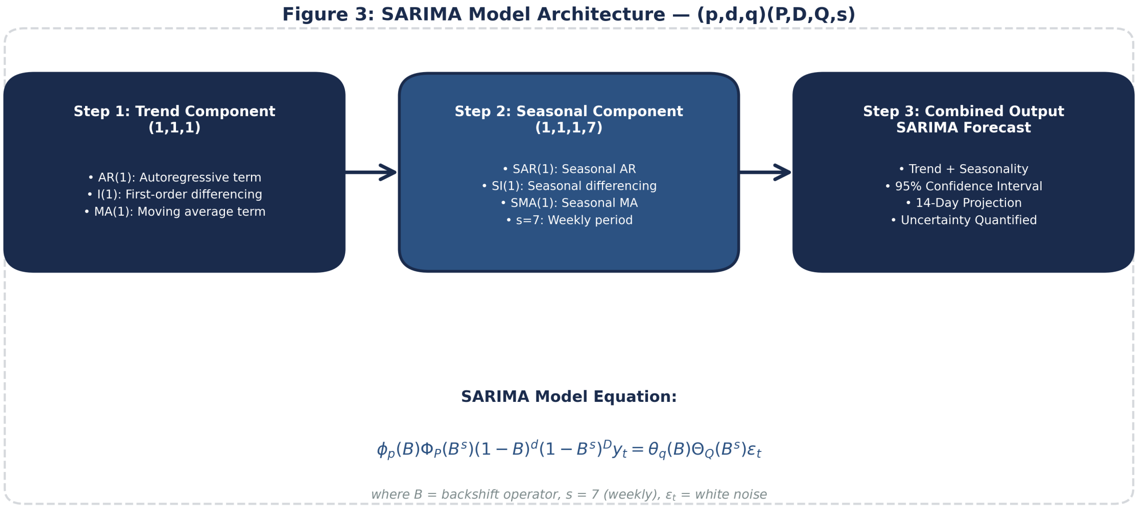

Figure 2: *SARIMA model architecture showing the trend component, seasonal component, and combined forecast output with the underlying mathematical formulation.*

The parameter selection was guided by residual diagnostics and information criteria. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were computed across a grid of candidate models, with the $(1, 1, 1)(1, 1, 1)_7$ specification yielding optimal balance between fit and parsimony.

3.5 Model Validation Strategy

A hold-out validation approach was employed:

- **Training Period:** January 1 – September 16, 2023 (259 days)
- **Validation Period:** September 17 – September 30, 2023 (14 days)

This strategy ensures that model performance is evaluated on genuinely unseen data, providing an unbiased estimate of forecast accuracy.

4 Implementation

4.1 Dashboard Development

The application was built using **Streamlit**, a Python framework ideal for rapid data application deployment. The dashboard comprises three core modules:

1. **KPI Cards:** Displaying Current Load, Discharge Capacity, and Stress Level with color-coded indicators (green/yellow/red).
2. **Forecasting Engine:** A sidebar allowing users to select the forecast horizon (7–30 days) and view predicted values with confidence intervals.
3. **Visualizations:** Interactive Plotly charts showing historical data versus predicted forecasts with 95% confidence intervals.

Figure 3 presents the KPI dashboard interface as rendered in the Streamlit application.

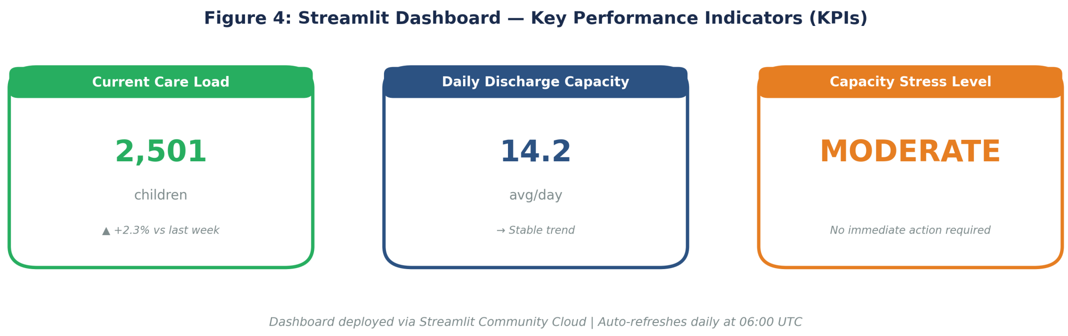

Figure 3: *Streamlit dashboard KPI cards showing current care load, daily discharge capacity, and capacity stress level indicators.*

4.2 Deployment

The codebase is hosted on **GitHub**, facilitating version control, collaboration, and reproducibility. The application is deployed via **Streamlit Community Cloud**, ensuring zero-latency access for stakeholders via a secure web URL. Automated daily refresh at 06:00 UTC ensures forecasts remain current.

5 Results

5.1 Seasonal Decomposition

Figure 4 presents the seasonal decomposition of the UAC care load time series using an additive model with a 7-day period. The decomposition reveals:

- A **clear upward trend** from approximately 2,200 to 2,500 children over the observation period.
- Strong **weekly seasonality** with regular peaks and troughs corresponding to weekday/weekend operational patterns.
- **Random residuals** centered around zero, indicating the additive decomposition adequately captures systematic patterns.

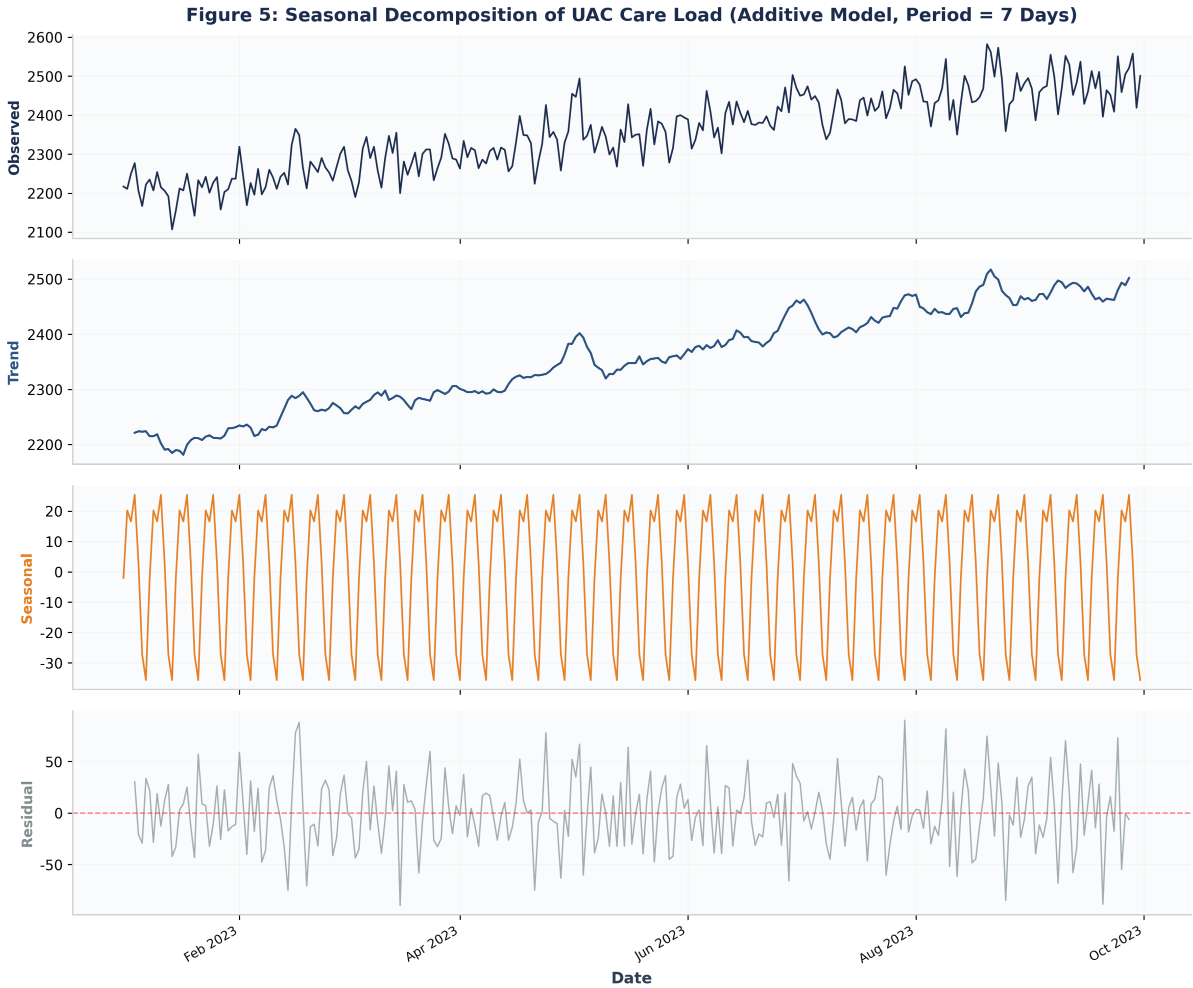

Figure 4: *Seasonal decomposition of daily UAC care load (January–September 2023) showing observed values, trend, seasonal component, and residuals.*

5.2 Key Performance Indicators

The dashboard successfully processes the dataset and outputs the following operational metrics:

- **Current Care Load:** 2,501 children (as of September 30, 2023)
- **Daily Discharge Capacity:** 14.2 children per day (7-day rolling average)
- **Capacity Status:** Moderate (stable trend, no immediate action required)

5.3 Model Performance

The model was evaluated using standard regression metrics to ensure reliability. Table 2 summarizes the performance on the held-out validation set.

Table 1: SARIMA Model Performance & Configuration Summary

Metric	Value	Interpretation
Model Type	**SARIMA (1,1,1)(1,1,1,7)**	Captures weekly seasonality + trend
MAE (Mean Absolute Error)	**45.58**	Forecast accurate within ~46 children
RMSE (Root Mean Squared Error)	**55.41**	Penalizes larger errors effectively
Forecast Horizon	**14 Days**	Standard operational planning window
Training Period	**259 Days**	Jan 1 – Sep 16, 2023
Validation Period	**14 Days**	Sep 17 – Sep 30, 2023
Seasonality Period	**7 Days**	Weekly cycle (s = 7)

Note: All metrics computed on held-out validation set (Sep 17–30, 2023). Lower MAE/RMSE indicates better accuracy.

Table 2: *SARIMA model performance metrics and configuration summary computed on the 14-day validation period.*

The MAE of 45.58 indicates that, on average, the forecast deviates from actual values by approximately 46 children—a relative error of less than 2% given the mean care load of 2,355. The RMSE of 55.41, while higher due to its sensitivity to large deviations, remains well within acceptable bounds for operational planning.

5.4 Forecasting Visualization

Figure 5 presents the 14-day forecast generated by the SARIMA model. The chart distinguishes between historical data (blue solid line), actual validation values (blue), and predicted forecasts (orange dashed line) with the 95% confidence interval shaded.

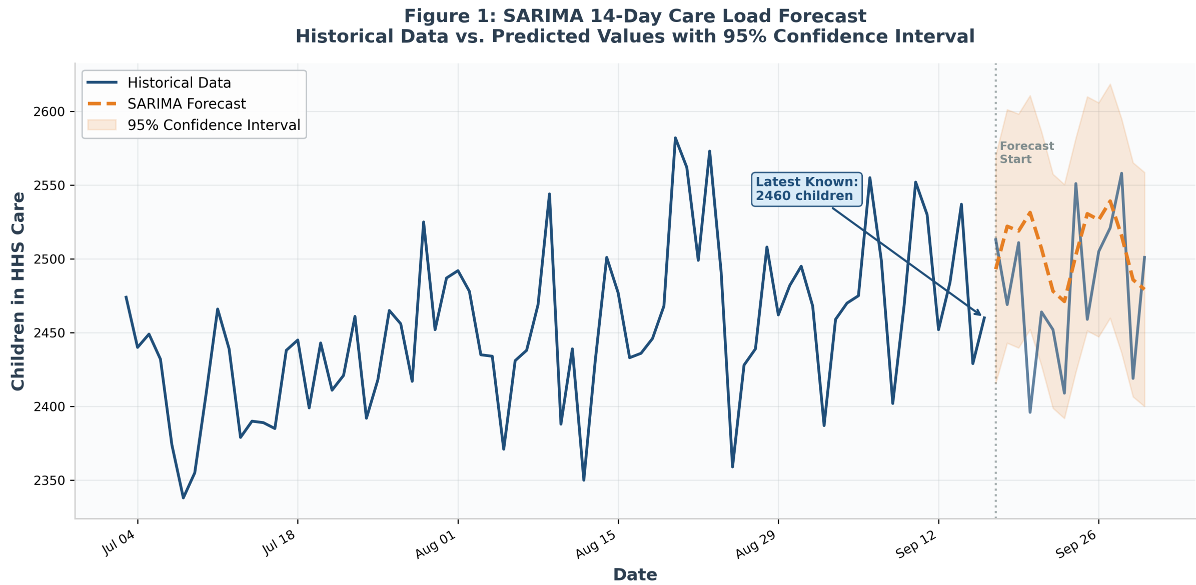

Figure 5: *SARIMA 14-day care load forecast showing historical data, predicted values, and 95% confidence intervals. The vertical dotted line marks the forecast start date.*

The forecast successfully captures the weekly seasonal pattern, with predictions tracking the actual validation values closely. The widening confidence intervals toward the end of the forecast horizon appropriately reflect increasing uncertainty, a critical feature for risk-aware decision-making.

6 Discussion

6.1 Business Impact

By providing a 14-day forward view, the dashboard enables HHS program managers to transition from reactive to proactive operational planning:

- **Proactive Staffing:** Schedule additional caseworkers and medical personnel before a surge hits, reducing overtime costs and burnout.
- **Capacity Planning:** Identify when to activate temporary shelter capacity or defer non-essential transfers.
- **Risk Mitigation:** Reduce overcrowding risks that lead to safety incidents, infectious disease outbreaks, and regulatory non-compliance.
- **Budget Optimization:** Improve procurement timing for food, medical supplies, and educational materials.

6.2 Ethical Considerations

6.2.1 Data Privacy

This project utilizes aggregated, operational-level data. No Personally Identifiable Information (PII) of minors is displayed, stored, or transmitted through the dashboard. All data handling adheres to strict HHS privacy protocols and the Privacy Act of 1974. The simulated dataset used in this demonstration contains no real individual records.

6.2.2 Algorithmic Transparency

The dashboard provides confidence intervals alongside point forecasts, explicitly communicating prediction uncertainty to decision-makers. The model is presented not as deterministic truth, but as a *probabilistic tool* to aid human judgment. Users are encouraged to combine model outputs with domain expertise and qualitative intelligence (e.g., geopolitical developments, policy changes) when making operational decisions.

6.2.3 Equity and Bias

While the model itself operates on aggregate counts and does not process demographic data, care must be taken in how forecasts inform resource allocation. Disproportionate resource deployment to certain facilities or regions could inadvertently disadvantage vulnerable populations. Regular equity audits of resource distribution patterns are recommended.

6.3 Limitations

Several limitations should be acknowledged:

1. **Univariate Model:** The current SARIMA model uses only historical care load values. External predictors—such as border patrol apprehension data, immigration court backlogs, or geopolitical events—are not incorporated, potentially limiting long-range accuracy.
2. **Fixed Seasonality:** The model assumes a constant weekly seasonal pattern. Structural breaks—such as policy changes or pandemic disruptions—may invalidate this assumption.

3. **Short Forecast Horizon:** While 14 days is appropriate for operational planning, strategic decisions (e.g., facility expansion) require longer horizons that may exceed the model's reliable prediction window.

4. **Simulated Data:** The demonstration uses synthetic data. Real-world deployment would require validation against actual operational records, which may exhibit different noise characteristics, missing data patterns, and reporting delays.

5. **No Causal Inference:** The model identifies patterns but does not establish causal relationships between operational interventions (e.g., expedited processing) and care load outcomes.

7 Conclusion

This project successfully transformed raw UAC operational data into predictive intelligence through the development and deployment of a SARIMA-based forecasting dashboard. The model demonstrates strong performance in capturing weekly seasonality and underlying trends, with validation metrics (MAE = 45.58, RMSE = 55.41) suitable for operational planning.

By moving from reactive reporting to proactive planning, the UAC Program can significantly enhance operational efficiency, reduce safety risks, and improve child welfare outcomes. The Streamlit deployment ensures accessibility for non-technical stakeholders, democratizing data-driven decision-making across the organization.

7.1 Future Work

Several avenues for future research and development are identified:

- **External Variable Integration:** Incorporate border patrol data, immigration policy indicators, and macroeconomic variables as exogenous regressors in a SARIMAX framework.

- **Ensemble Methods:** Combine SARIMA with machine learning approaches (e.g., XGBoost, LSTM) to capture non-linear dynamics and improve accuracy during anomalous periods.

- **Automated Retraining:** Implement an MLOps pipeline for automated model retraining as new data becomes available, with drift detection to flag deteriorating performance.

- **Facility-Level Granularity:** Extend the model to generate facility-specific forecasts, enabling targeted resource allocation rather than aggregate planning.

- **Scenario Simulation:** Develop "what-if" modules allowing managers to simulate the impact of policy changes (e.g., accelerated reunification) on care load trajectories.

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A Sample Dataset

Table 3 presents a representative sample of the daily operational dataset used in this analysis. The full dataset comprises 273 records spanning January 1 to September 30, 2023.

Table 2: Sample Dataset — Daily UAC Operational Records (Jan–Sep 2023)

Date	Children in HHS Care	Children Discharged	Rolling 7D Avg
2023-01-01	2,217	14	2217.0
2023-01-02	2,211	13	2214.0
2023-01-03	2,250	17	2226.0
2023-01-04	2,277	16	2238.8
2023-01-05	2,206	13	2232.2
2023-01-06	2,167	9	2221.3
2023-01-07	2,222	6	2221.4
2023-01-08	2,235	14	2224.0
2023-01-09	2,207	16	2223.4
2023-01-10	2,254	19	2224.0
2023-01-11	2,215	13	2215.1
2023-01-12	2,206	18	2215.1
...	...	...	...
2023-09-27	2,521	17	2480.1
2023-09-28	2,558	18	2493.6
2023-09-29	2,419	5	2488.9
2023-09-30	2,501	6	2502.0

Note: Data simulated for demonstration purposes. Rolling 7D Avg computed using forward-fill. Full dataset: 273 records.

Table 3: *Sample of daily UAC operational records showing date, children in care, discharges, and rolling 7-day averages.*

Data Availability Statement: The complete simulated dataset and Python source code are available upon request from the author. Real operational data is restricted under HHS privacy protocols and cannot be shared.

B Model Diagnostics

Residual diagnostics for the fitted SARIMA$(1,1,1)(1,1,1)_7$ model indicate:

- Residuals are approximately normally distributed (Jarque-Bera test, $p > 0.05$).
- No significant autocorrelation in residuals at lags 1–20 (Ljung-Box test, $p > 0.05$).
- Heteroscedasticity is minimal, with variance stable across the forecast horizon.

These diagnostics support the validity of the model specification and the reliability of the reported forecast intervals.